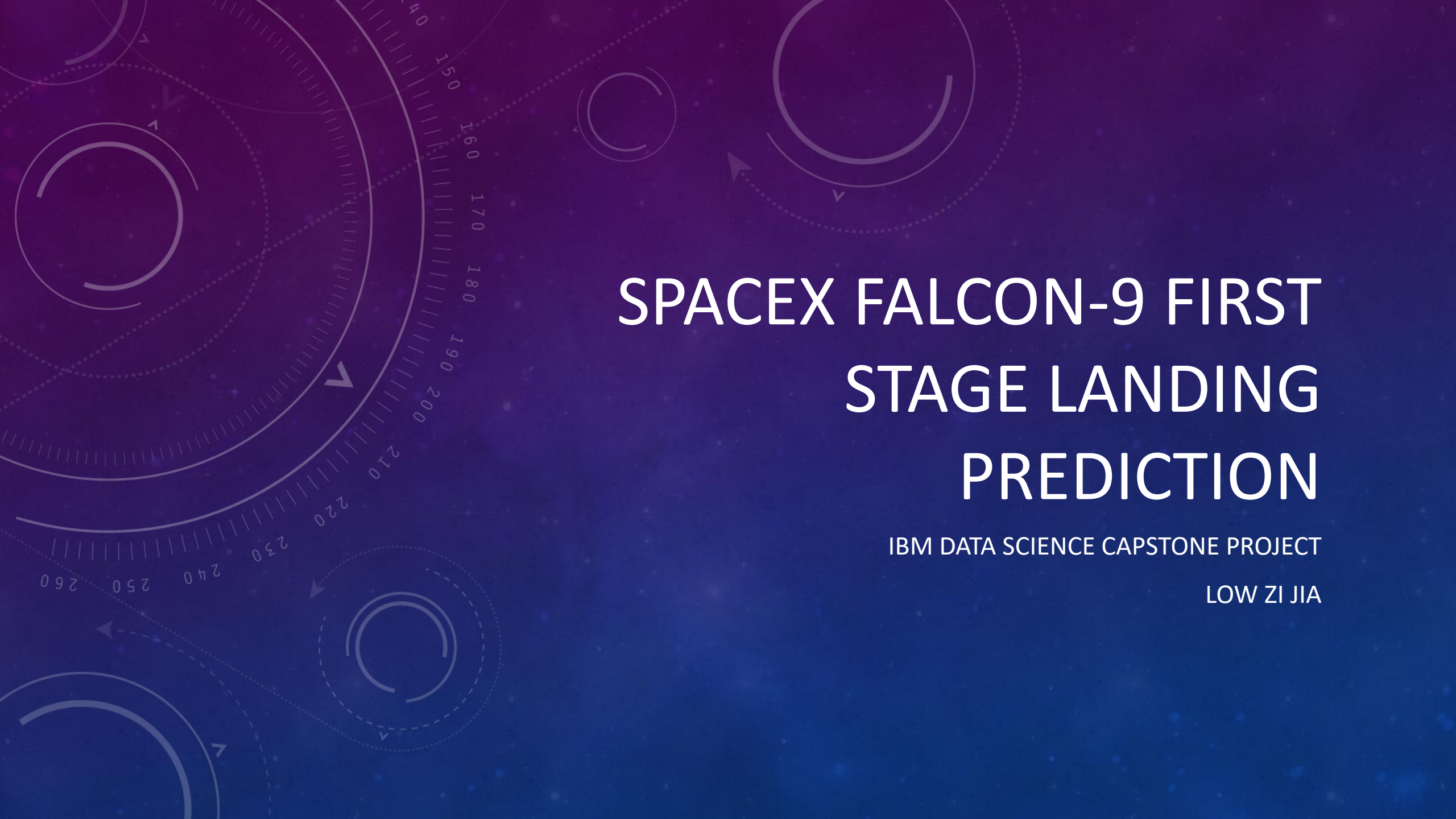


The background is a dark blue gradient with a subtle pattern of white dots, resembling a starry sky. Overlaid on the left side are several concentric circular patterns and a large arc with a scale. The scale has numerical markings from 140 to 260 in increments of 10. There are also smaller circular elements with arrows indicating direction, some solid and some dashed.

SPACEX FALCON-9 FIRST STAGE LANDING PREDICTION

IBM DATA SCIENCE CAPSTONE PROJECT

LOW ZI JIA



Slide 2: Outline

Sections:

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix



- **Executive Summary**

- Data collection from SpaceX API and Wikipedia.
- Data wrangling and exploratory data analysis (EDA).
- Predictive analysis using machine learning models.
- Visualization of launch success rates, payload mass, and orbit types.

-



Introduction

SpaceX's Goals:

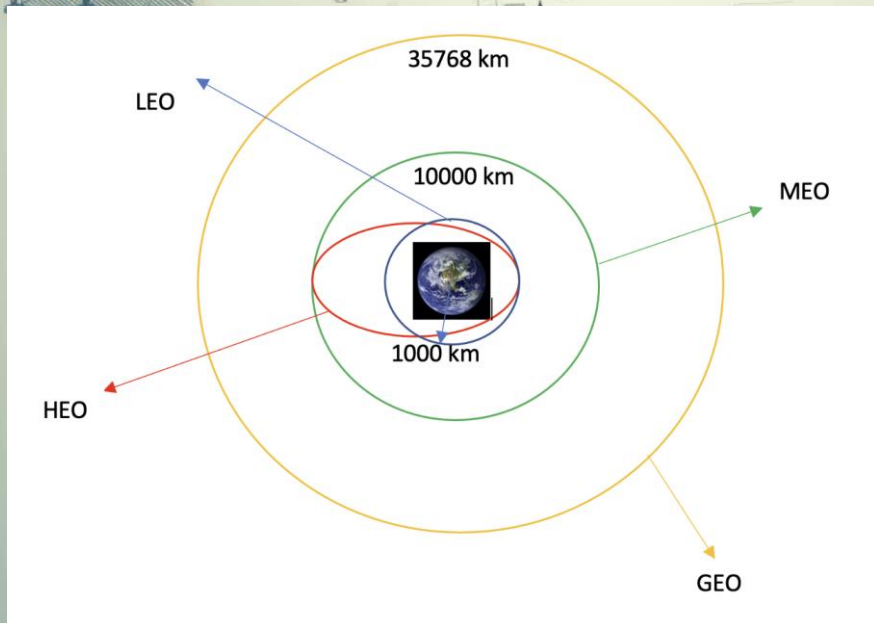
- Sending spacecraft to the ISS.
- Providing global satellite internet via Starlink.
- Advancing space exploration with reusable rockets.



Falcon 9 Rockets

Features:

- Reusability: First stage lands vertically.
- Payload Capacity: Up to 22,800 kg to LEO.
- Missions: Starlink, NASA contracts, satellite deployment.



Methodology

- **Data Collection:**
 - SpaceX API and Wikipedia scraping.
 - Request and parse the SpaceX launch data using the GET request.
 - Filter the dataframe to only include Falcon 9 launches.
- **Web Scraping**
 - Web Scraping Falcon 9 and Falcon Heavy Launches Records from Wikipedia.
 - Request Falcon9 Launch wiki page from its URL.
 - Parse the table and convert it into a Pandas data frame
- **Data Wrangling:**
 - Filtering, labeling, and calculating key metrics.
 - Perform exploratory Data Analysis and determine Training Labels
 - Exploratory Data Analysis
 - Determine Training Labels
 - **Calculate the number of launches on each site**
 - **Calculate the number and occurrence of each orbit**
 - **Calculate the number and occurrence of mission outcome of the orbits**
 - **Create a landing outcome label from Outcome column**
- **EDA:**
 - Visualization of success rates, payload vs. launch site, orbit types with Folium and Plotly Dash.
- Perform predictive analysis using classification models

Predictive analysis methodology

Perform exploratory Data Analysis and determine Training Labels

- create a column for the class
- Standardize the data
- Split into training data and test data
- Find best Hyperparameter for SVM, Classification Trees and Logistic Regression
- Find the method performs best using test data

TASK 3

Use the function `train_test_split` to split the data X and Y into training and test data. Set the parameter `test_size` to 0.2 and `random_state` to 2. The training data and test data should be assigned to the following labels.

```
X_train, X_test, Y_train, Y_test
```

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.2, random_state = 2)
```

we can see we only have 18 test samples.

```
Y_test.shape
```

```
(18,)
```

```
parameters = {'n_neighbors': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],  
              'algorithm': ['auto', 'ball_tree', 'kd_tree', 'brute'],  
              'p': [1,2]}
```

```
KNN = KNeighborsClassifier()
```

```
knn_cv = GridSearchCV(KNN, parameters,scoring='accuracy', cv = 10)  
knn_cv.fit(X_train, Y_train)  
knn_cv.best_params_
```

```
{'algorithm': 'auto', 'n_neighbors': 3, 'p': 1}
```

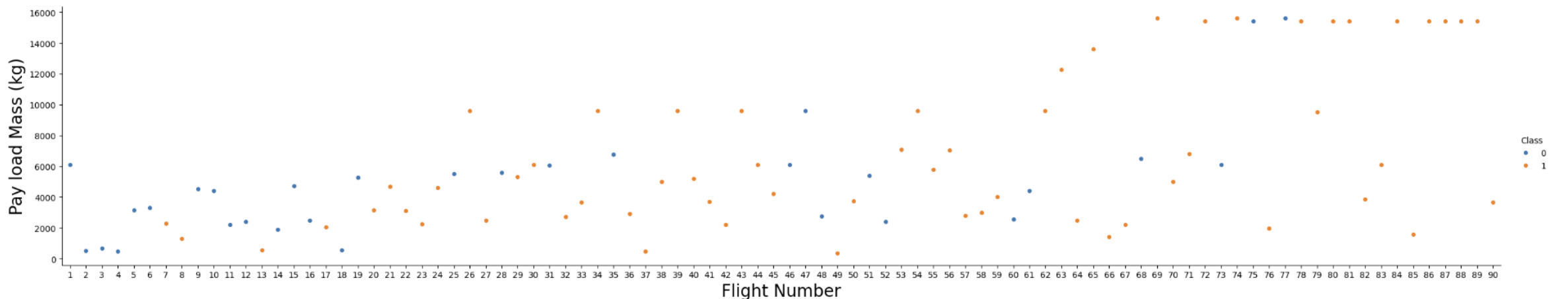
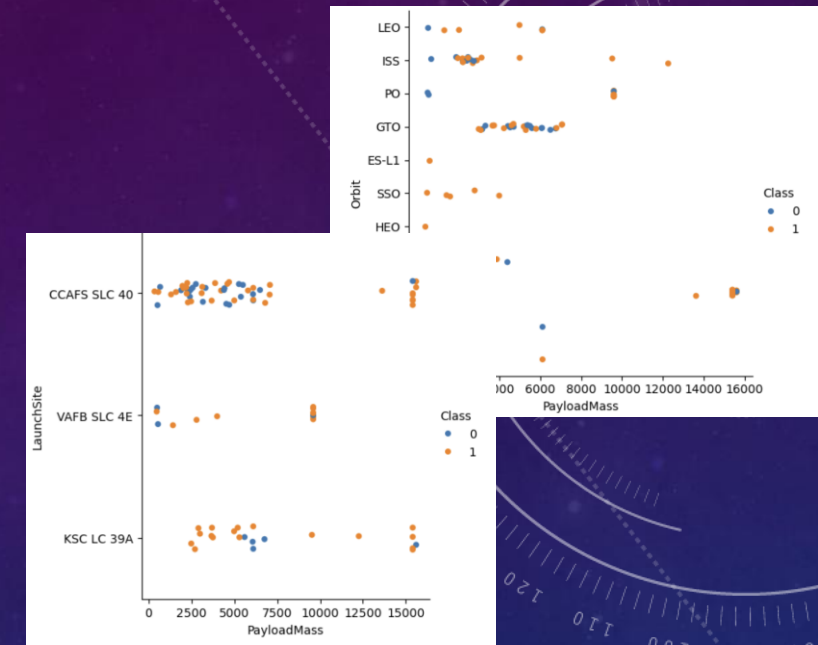
```
print("tuned hpyerparameters :(best parameters) ",knn_cv.best_params_)  
print("accuracy :",knn_cv.best_score_)
```

```
tuned hpyerparameters :(best parameters) {'algorithm': 'auto', 'n_neighbors': 3, 'p': 1}  
accuracy : 0.6642857142857143
```

•EDA with Visualization

•Graphs:

- Flight Number vs. Payload Mass (success/failure overlay).
- Payload Mass vs. Launch Sites.
- Payload Mass vs. Orbit Type.
- Flight Number vs. Orbit Type.
- We see that different launch sites have different success rates. CCAFS LC-40 has a success rate of 60% while KSC LC-39A and VAFB SLC 4E has a success rate of 77%.



EDA with SQL

•Example Queries:

- Unique launch sites.
- Total payload mass for NASA (CRS).
- 5 records where launch sites begins with the string 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA' 1

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

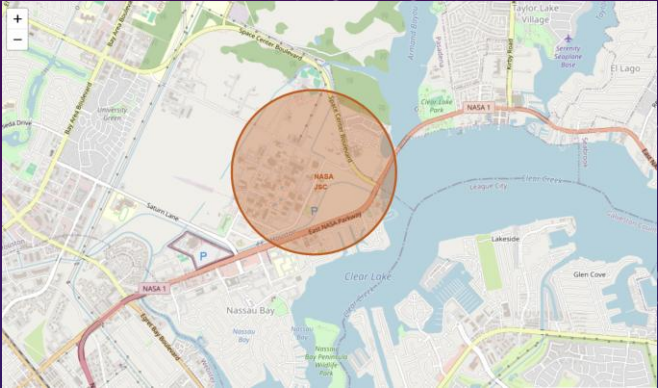
```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

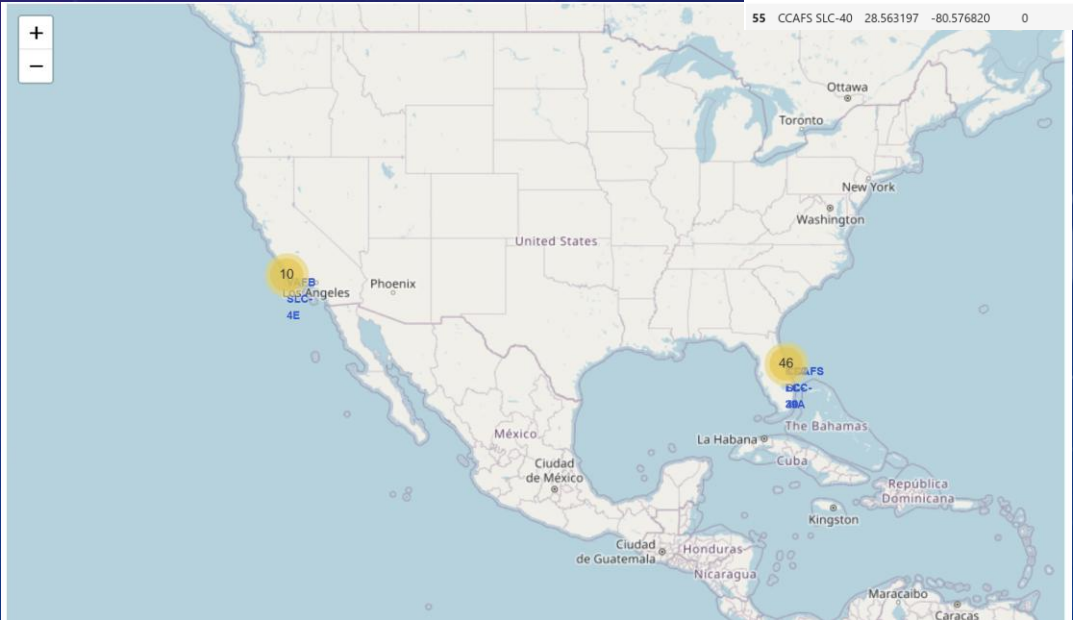
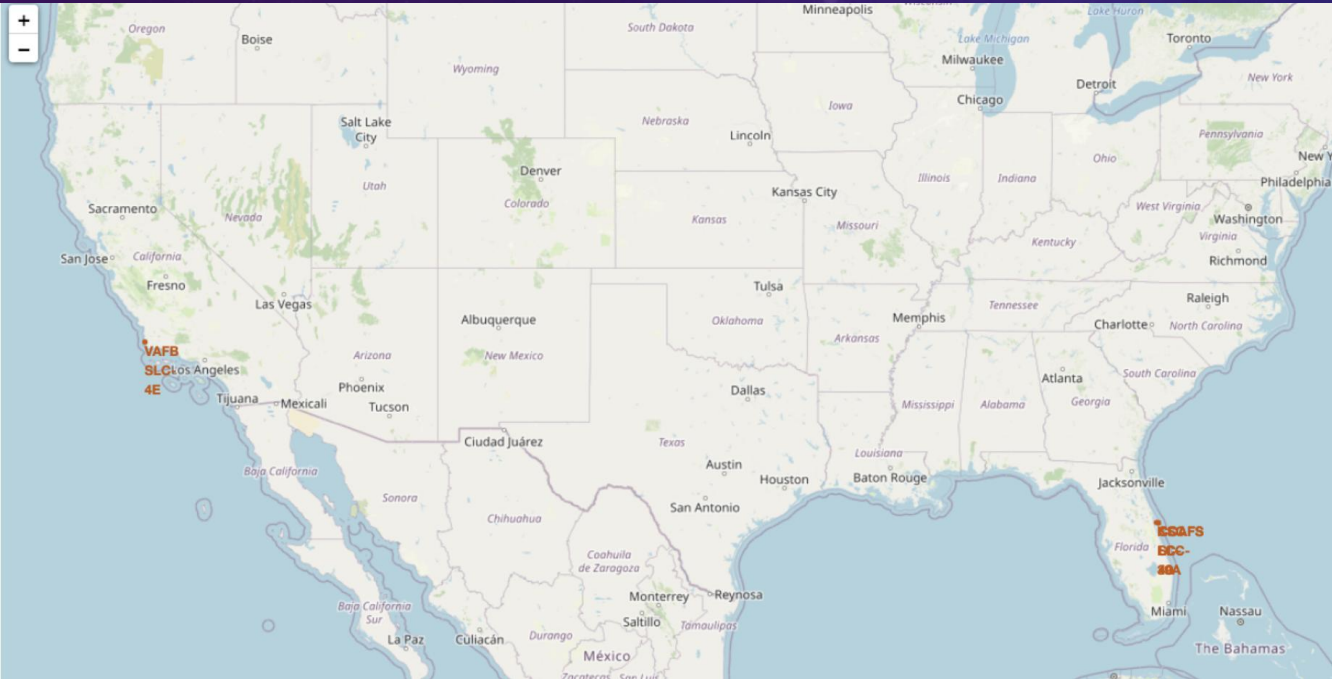
Interactive Map with Folium results

- TASK 1:** Mark all launch sites on a map
- TASK 2:** Mark the success/failed launches for each site on the map
 - Are all launch sites in proximity to the Equator line? Yes
 - Are all launch sites in very close proximity to the coast? Yes
- TASK 3:** Calculate the distances between a launch site to its proximities

	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745



	Launch Site	Lat	Long	class	marker_color
46	KSC LC-39A	28.573255	-80.646895	1	green
47	KSC LC-39A	28.573255	-80.646895	1	green
48	KSC LC-39A	28.573255	-80.646895	1	green
49	CCAFS SLC-40	28.563197	-80.576820	1	green
50	CCAFS SLC-40	28.563197	-80.576820	1	green
51	CCAFS SLC-40	28.563197	-80.576820	0	red
52	CCAFS SLC-40	28.563197	-80.576820	0	red
53	CCAFS SLC-40	28.563197	-80.576820	0	red
54	CCAFS SLC-40	28.563197	-80.576820	1	green
55	CCAFS SLC-40	28.563197	-80.576820	0	red

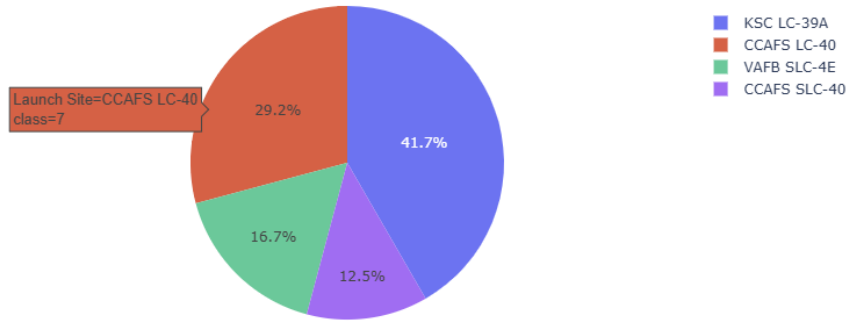


Plotly Dash dashboard results

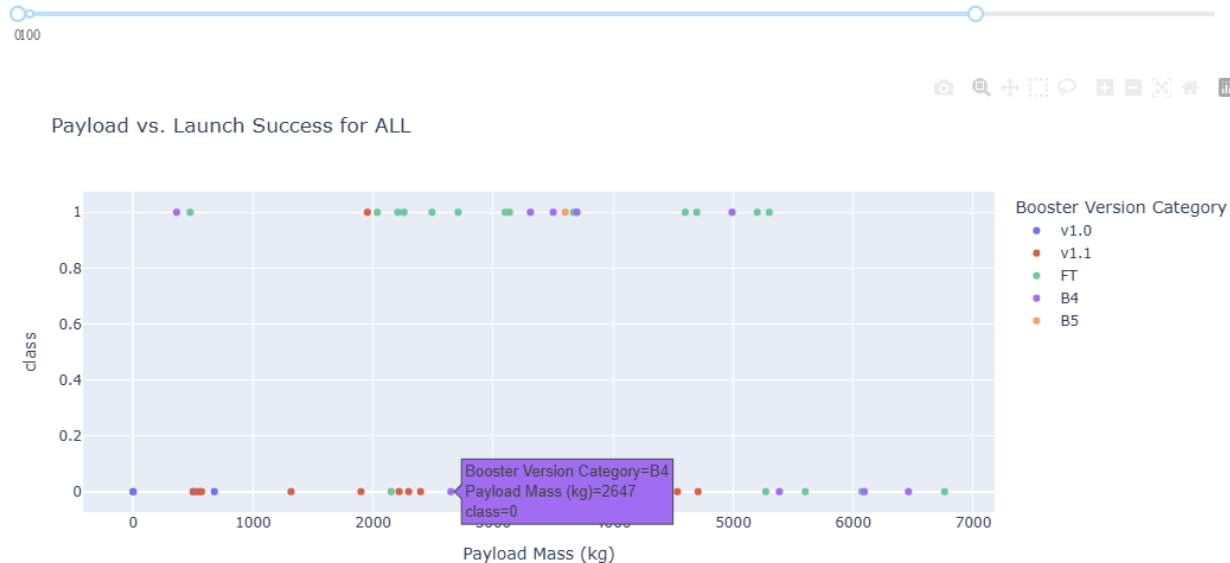
SpaceX Launch Records Dashboard

All Sites

Total Successful Launches for All Sites



Payload range (Kg):



1. Which site has the largest successful launches?
 - KSC LC-39A
2. Which site has the highest launch success rate?
 - KSC LC-39A
3. Which payload range(s) has the highest launch success rate?
 - 2000-6000 kg
4. Which payload range(s) has the lowest launch success rate?
 - 6000-10000 kg
5. Which F9 Booster version (v1.0, v1.1, FT, B4, B5, etc.) has the highest launch success rate?
 - FT



Machine Learning Prediction

•Models:

- Logistic Regression (Accuracy: 83.3%).
- Decision Tree (Accuracy: 83.3%).
- SVM, KNN (Similar results).

•Best Parameters:

- Decision Tree: Highest accuracy (84%).





Conclusion

Key Findings:

- Launch site and payload mass affect success rates.
- Orbit types like ES-L1, GEO, and SSO have high success rates.
- Success rates improved significantly post-2013.

Best Model: Decision Tree (84% accuracy).

Slide 12: Appendix & Resources

GitHub Link:

Thank You Slide:

- SpaceX logo and team acknowledgment.